# Vancomycin vs Daptomycin For Inpatient MRSA Infections: A Comprehensive Evidence-Based Comparison

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# Objectives

- Compare clinical efficacy between daptomycin and vancomycin
- Analyze failure and mortality between both agents
- Assess safety profiles
- Review cost analysis data between both agents

# Executive Summary

**Key Points:**

- **Superior Efficacy:** Daptomycin shown to reduce clinical failure rate (29% vs 45%, p= 0.007).[1]
- **Improved Survival:** lower 30 day mortality rate (6.1% vs 15.3%, p = 0.024).[1]
- **Improved Safety**: Less severe adverse events (RR 0.57, 95% CI 0.36-0.90)[2]
- **Cost-Effective:** Total healthcare savings despite higher drug acquisition costs.[3]
- **Early Switching is Critical**: 52% reduction in mortality when switched within 3 days (HR 0.48).[4]

# Clinical Dilemma

## Why Vancomycin May Be Failing Our Patients:

### Treatment Failure Rates:

- Vancomycin is associated with 45-48% clinical failure independent of MIC
- ICU patients are more vulnerable

### The MIC Creep Phenomenon:

- Consistent increase in vancomycin MIC over time
- 39% of MRSA isolates now have MIC >1 microgram/mL
- Associated with worsening outcomes

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Murray KP, Zhao JJ, Davis SL, et al. Early use of daptomycin versus vancomycin for methicillin-resistant Staphylococcus aureus bacteremia with vancomycin minimum inhibitory concentration >1 mg/L: a matched cohort study. Clin Infect Dis. 2013;56(11):1562-1569. doi:10.1093/cid/cit112
Claeys KC, Zasowski EJ, Casapao AM, et al. Daptomycin improves outcomes regardless of vancomycin MIC in a propensity-matched analysis of methicillin-resistant Staphylococcus aureus bloodstream infections. Antimicrob Agents Chemother. 2016;60(10):5841-5848. doi:10.1128/aac.00227-16

# Head-to-Head Randomized Controlled Trial

**Landmark RCT: Fowler et al. NEJM 2006**

**Study Design:**

- Randomized, open-label trial
- Daptomycin 6 mg/kg daily vs standard therapy (vancomycin or antistaphylococcal penicillin + gentamicin)
- S. aureus bacteremia and endocarditis

**Primary Results:**

- Non Inferiority demonstrated at 42 days post-therapy
- Daptomycin showed 44.4% success vs 41.7% success with standard therapy, absolute difference of 2.4% (95% CI: -8.0 to 12.8)

**Limitations:**

- Wide confidence intervals with low success rates in both arms
- Microbiological failure occurred more with daptomycin, not statistically significant.
- Only 18 patients with left sided endocarditis enroled (9 in each arm)
- 6 mg/kg dose may be suboptimal for serious infections
- Comparator arm design- addition of low dose gentamicin



Fowler VG Jr, Boucher HW, Corey GR, et al. Daptomycin versus standard therapy for bacteremia and endocarditis caused by *Staphylococcus aureus*. *N Engl J Med*. 2006;355(7):653-665. doi:10.1056/NEJMoa053783

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**Clinical Implication:**

- Despite limitations, study still shows that daptomycin is a viable option for S. aureus bacteremia and right-sided endocarditis particularly in patients who cannot tolerate vancomycin.
- Modest success rates further re-enforces that antibiotic therapy alone is insufficient, adequate source control and surgical intervention are critical for optimal outcome



Fowler VG Jr, Boucher HW, Corey GR, et al. Daptomycin versus standard therapy for bacteremia and endocarditis caused by *Staphylococcus aureus*. *N Engl J Med*. 2006;355(7):653-665. doi:10.1056/NEJMoa053783

# Meta-Analysis of RCTs

**Boulekbache et al., 2024**: 8 RCTs, 1095 Patients, publications through June 30th, 2021

## Study compositions:

- 75% skin/soft tissue infections, 12.5% bacteremia, 12.5% bone/joint infection

## Efficacy and Safety Outcomes:

| Outcome | Result | P-value or significance |
|---|---|---|
| All-cause mortality | No significant difference | >0.05 |
| Clinical cure | No significant difference | >0.05 |
| Microbiological cure | RR 1.17 (1.01-1.35) | Favors daptomycin |
| Severe adverse events (grade >3) | RR 0.57 (0.36-0.90) | 43% reduction with daptomycin |

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Boulekbache A, Maldonado F, Kavafian R, et al. Comparison of daptomycin and glycopeptide efficacy and safety for the treatment of Gram-positive infections: a systematic review and meta-analysis. J Antimicrob Chemother. 2024;79(4):712-721. doi:10.1093/jac/dkae026

# Meta-Analysis of RCTs

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## Limitations:

- 75% of included trials focused on skin and soft tissue infection, limits generalizability to serious infection
- Clinical heterogeneity concerns by combining multiple trials may result in inconsistent results and reduced generalizability

## Clinical Implications

- Supports daptomycin as a safe alternative to glycopeptide with comparable clinical efficacy and better microbiological eradication for gram positive infection
- However, the  author's recommendation is to approach findings with caution, further RCTs in bacteremia and endocarditis is needed to address evidence gap.



Boulekbache A, Maldonado F, Kavafian R, et al. Comparison of daptomycin and glycopeptide efficacy and safety for the treatment of Gram-positive infections: a systematic review and meta-analysis. J Antimicrob Chemother. 2024;79(4):712-721. doi:10.1093/jac/dkae026

# Real World Evidence: Retrospective Cohort Study

**Claeys et al. , 2016:** 262 Matched Patients, All MIC

**Primary Endpoint:** clinical failure defined as a composite 30-day mortality, bacteremia duration of over 7 days or change in treatment due to persistent or worsening symptoms.

**Findings:**

- **Clinical failure:** Vancomycin 45.0% vs Daptomycin 29.0% p = 0.007
- **30-Day All-Cause Mortality:** Vancomycin 15.3% vs Daptomycin 6.1%, p = 0.024
- No significant difference in bacteremia duration

**MIC independent benefit:** Daptomycin effective across all vancomycin MIC including MIC <1 mg/L

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Claeys KC, Zasowski EJ, Casapao AM, et al. Daptomycin improves outcomes regardless of vancomycin MIC in a propensity-matched analysis of methicillin-resistant Staphylococcus aureus bloodstream infections. Antimicrob Agents Chemother. 2016;60(10):5841-5848. doi:10.1128/aac.00227-16

# Real World Evidence: Cohort Study

**Claeys et al. , 2016:** 262 Matched Patients, All MIC

**Independent Predictors of Treatment Failure**:

- Vancomycin therapy (aOR 2.16, 95% CI 1.24-3.76)
- ICU admission (aOR 2.46, 95% CI 1.34-4.54)
- Infective endocarditis as primary source (aOR 2.33, 95% CI 1.16-4.68)

**Strength:**

- Propensity matching  to control confounding bias
- Clinically relevant composite endpoint

**Limitations:**

- Retrospective design
- No dosing details about vancomycin or achievement of trough levels
- No data on source control
- Included diverse source of bacteremia



Claeys KC, Zasowski EJ, Casapao AM, et al. Daptomycin improves outcomes regardless of vancomycin MIC in a propensity-matched analysis of methicillin-resistant Staphylococcus aureus bloodstream infections. Antimicrob Agents Chemother. 2016;60(10):5841-5848. doi:10.1128/aac.00227-16

# High Vancomycin MIC Patients

**Murray et al., 2013:** 170 Matched Patients with MRSA bacteremia, Vancomycin MIC >1 microgram/mL from January 2006 - December 2010

## Outcomes:

- **Clinical Failure (30-day mortality or persistent bacteremia >7 days):**
  - Vancomycin 48.2% vs Daptomycin 20.0% ($p < 0.001$)
- **30- Day Mortality:**
  - Vancomycin 12.9% vs Daptomycin 3.5% ($p = 0.047$)
- **Persistent Bacteremia:**
  - Vancomycin 42.4% vs Daptomycin 18.8% ($p = 0.001$)
- **Meta-Analysis (Samura et al., 2022)**: 907 Patients, MIC > 1 microgram/mL
  - **Mortality**: OR 0.53 (95% CI 0.29-0.98) favoring daptomycin
  - **Treatment success:** OR 2.20 (95% CI 1.63-2.96) favoring daptomycin
  - Greatest benefit with bone/joint infections



Murray KP, Zhao JJ, Davis SL, et al. Early use of daptomycin versus vancomycin for methicillin-resistant Staphylococcus aureus bacteremia with vancomycin minimum inhibitory concentration >1 mg/L: a matched cohort study. Clin Infect Dis. 2013;56(11):1562-1569. doi:10.1093/cid/cit112

# Timing Is Important: Early Switching

**Schweizer et al. 2021**: 124 VA hospital, 7,411 Patients

**Early Switch <3 days:**

- 30 day mortality HR 0.48 (95% CI 0.25-0.92)
- 52% mortality reduction

**Late Switch (any time during hospitalization):**

- 30 day mortality HR 0.87 (95% CI 0.69-1.09)
- No significant benefit

**Clinical Implication:**

- The window of opportunity is narrow, benefit shown with switching within 3 days to achieve mortality benefit.



Schweizer ML, Richardson K, Vaughan Sarrazin MS, et al. Comparative effectiveness of switching to daptomycin versus remaining on vancomycin among patients with methicillin-resistant Staphylococcus aureus (MRSA) bloodstream infections. Clin Infect Dis. 2021;72(suppl 1):S68-S73. doi:10.1093/cid/ciaa1572

# Safety Comparison - Nephrotoxicity

**Meta-Analysis (Zhang et al., 2023)**: 24 Studies

- Adverse events: OR 3.21 (95% CI 1.43 -7.19)
- Vancomycin has 3.2 fold higher risk of adverse events compared to daptomycin

**Guadard et al., 2019**: Cardiovascular ICU Patients

**Acute Kidney Injury at Day 7:**

- Vancomycin: 64% (28/44) vs Daptomycin 21% (6/28); p= 0.001

**Renal Replacement Therapy Required:**

- Vancomycin 30% (13/44) vs Daptomycin 7% (2/28); p = 0.035

**Clinical Impact:** Vancomycin was the only independent predictor of AKI in this study



Zhang G, Zhang N, Xu J, Yang T, Yin H, Cai Y. Efficacy and safety of vancomycin for the treatment of Staphylococcus aureus bacteraemia: a systematic review and meta-analysis. Int J Antimicrob Agents. 2023;62(4):106946. doi:10.1016/j.ijantimicag.2023.106946

# Safety Comparison - Monitoring Burden

| Parameter | Vancomycin | Daptomycin |
|---|---|---|
| Therapeutic Drug Monitoring (TDM) | Required (AUC goal of 400-600 or trough 15-20 mg/L) | Not required |
| Frequency of Labs | 2-3 times/week minimum | Weekly CPK only |
| Dose Adjustments | Frequent, based on levels and renal function | weight based, stable makes it appropriate for OPAT |
| Pharmacist Time | High (TDM, dose adjustment, charting) | Low (no TDM) |
| Nursing Time | High (frequent blood draws) | Low (weekly/monitoring) |
| Risk of high drug levels | Significant, especially with OPAT | Minimal |

# Cost Effective Analysis - ASVE Cost of Acquisition

**Vancomycin:**

- 500 mg = $7.25 per vial
- 750 mg = $8.19 per vial
- 1 gram = $14.53 per vial
- 10 gram = $16.40 per vial

**Daptomycin:**

- 500 mg vial = $15.03

# Cost Effective Analysis - Impact of Length of Stay

**Browne et al., 2016:** UK Healthcare Cost Analysis for MRSA Bacteremia-Endocarditis

**Equal Length of Stay:**

- Daptomycin: £17,917 per patient
- Vancomycin: £ 17,165 per patient
- Cost difference of £752, more expensive with daptomycin

**Reduced Length of Stay (42 to 28 days with daptomycin)**

- Daptomycin saved £4037 per patient
- Reduction in hospital stay by 14 days
- Main saving drivers include fewer therapeutic changes and earlier discharge eligibility



Browne C, Muszbek N, Chapman R, et al. Comparative healthcare-associated costs of methicillin-resistant Staphylococcus aureus bacteraemia-infective endocarditis treated with either daptomycin or vancomycin. Int J Antimicrob Agents. 2016;47(5):357-361. doi:10.1016/j.ijantimicag.2016.02.006

# Hidden Costs With Vancomycin

- **Nephrotoxicity Management Costs:**
  - RRT requirement, cost of dialysis , extended ICU stay for AKI management, risk of progression to CKD
- **Therapeutic Drug Monitoring:**
  - Trough monitoring/AUC guided monitoring, lab cost (2-3/wk minimum), pharmacist time for dose adjustment, nursing time, additional blood draws



Norris AH, Shrestha NK, Allison GM, et al. 2018 Infectious Diseases Society of America clinical practice guideline for the management of outpatient parenteral antimicrobial therapy. Clin Infect Dis. 2019;68(1):e1-e35. doi:10.1093/cid/ciy745

# Hidden Costs With Vancomycin

- **Treatment Failure and Switch Costs**
  - Therapy changes, extended hospitalizations and additional interventions
- **OPAT (outpatient parenteral antimicrobial therapy) Challenges:**
  - Monitoring challenges, more adverse events due to increase toxicity risk
- **Resource Opportunity Cost:**
  - Free up pharmacy resources, nursing time

# Putting It All Together - When To Choose Daptomycin

**Strong Evidence for Daptomycin over Vancomycin**:

- **Vancomycin MIC >1 microgram/mL:**
  - Consider early switch even before MIC result if high suspicion
- **Compromised renal function or Acute Kidney Injury**
- **OPAT Candidates:**
  - No TDM required (weekly CPK only)
  - Less interruption in therapy
- **ICU Patients with MRSA Bacteremia:**
  - ICU admission shown to be an independent predictor of vancomycin failure
  - Early switch (≤3 days) reduces mortality by 52%
- **Complicated Bacteremia/Endocarditis:**
  - Infective endocarditis as primary source of bacteremia
  - High dose daptomycin (8-10 mg/kg) for left sided endocarditis

# Vancomycin May Still Be Appropriate

- Pneumonia (daptomycin inactivated by lung surfactant)
- CNS infections (Poor CNS penetration with daptomycin)
- Vancomycin MIC ≤1 microgram/mL with good clinical response (consider switch in AKI)

# Questions?

# References:


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2. Boulekbache A, Maldonado F, Kavafian R, et al. Comparison of daptomycin and glycopeptide efficacy and safety for the treatment of Gram-positive infections: a systematic review and meta-analysis. J Antimicrob Chemother. 2024;79(4):712-721. doi:10.1093/jac/dkae026
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4. Schweizer ML, Richardson K, Vaughan Sarrazin MS, et al. Comparative effectiveness of switching to daptomycin versus remaining on vancomycin among patients with methicillin-resistant Staphylococcus aureus (MRSA) bloodstream infections. Clin Infect Dis. 2021;72(suppl 1):S68-S73. doi:10.1093/cid/ciaa1572
5. Murray KP, Zhao JJ, Davis SL, et al. Early use of daptomycin versus vancomycin for methicillin-resistant Staphylococcus aureus bacteremia with vancomycin minimum inhibitory concentration >1 mg/L: a matched cohort study. Clin Infect Dis. 2013;56(11):1562-1569. doi:10.1093/cid/cit112
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